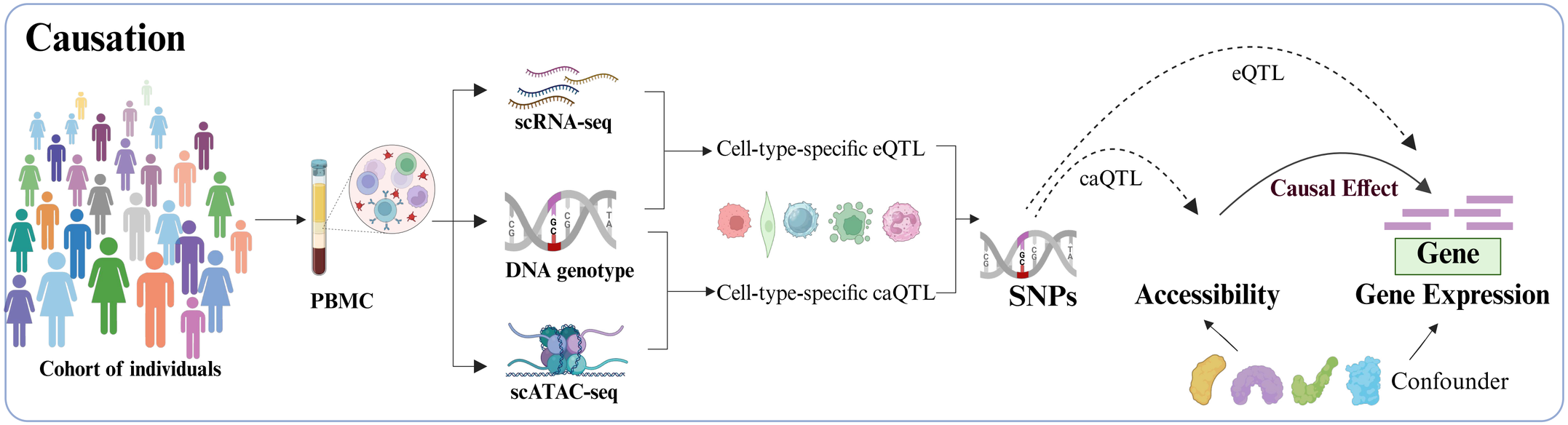
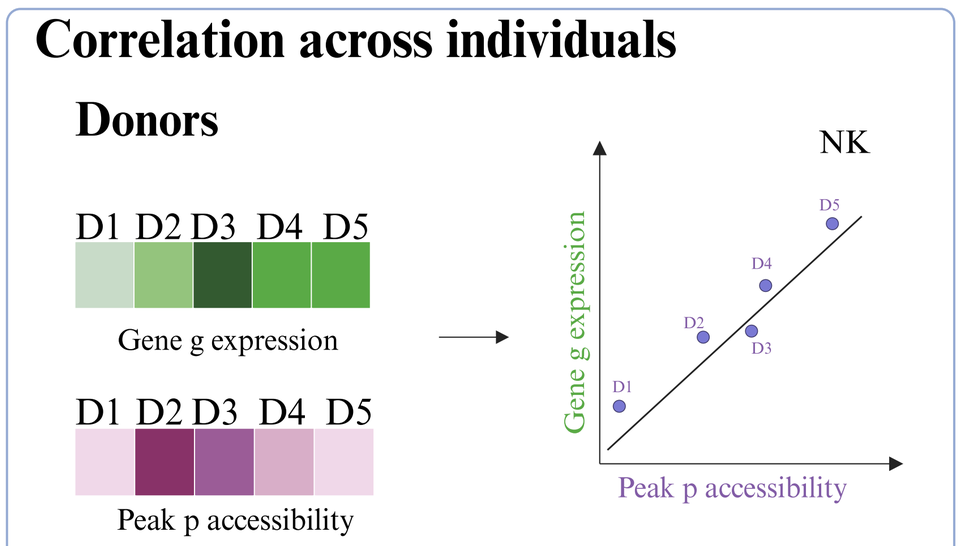
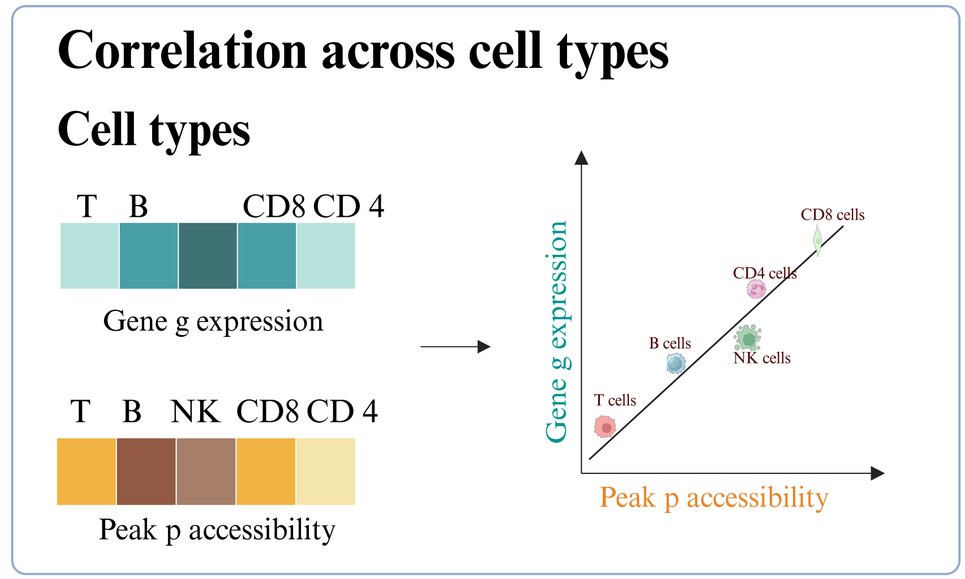
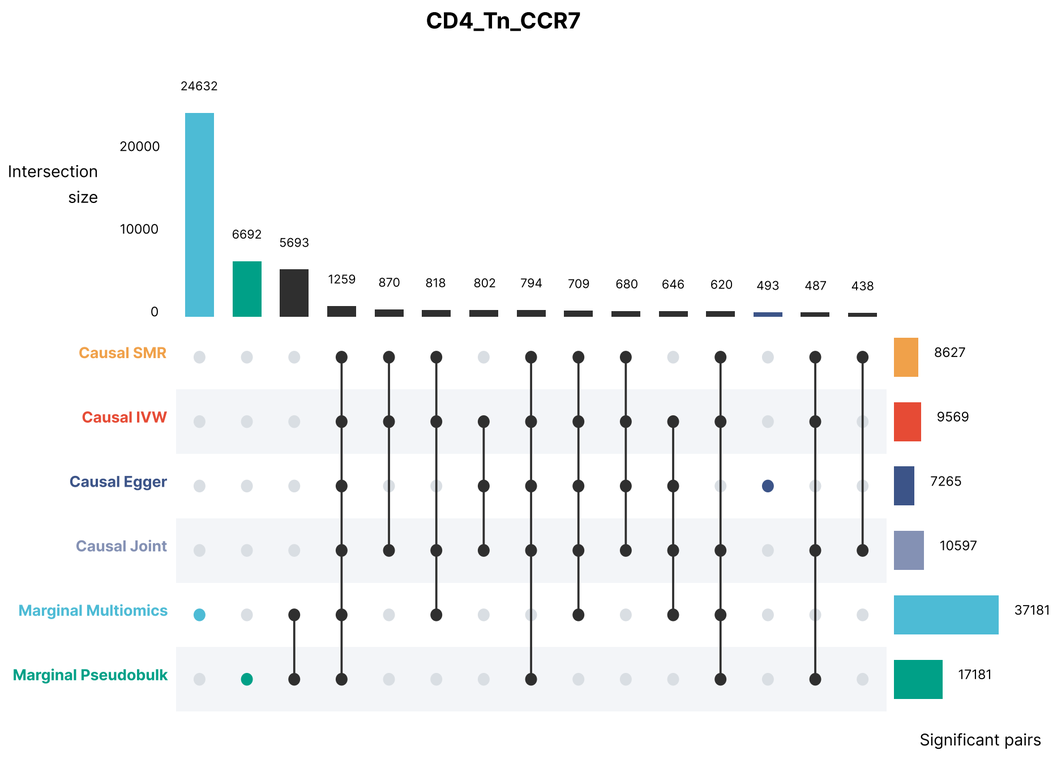
A


Causation
Cohort of individuals
PBMC
scRNA-seq
DNA genotype
scATAC-seq
Cell-type-specific eQTL
Cell-type-specific caQTL
SNPs
eQTL
caQTL
Causal Effect
Gene
Accessibility
Gene Expression
Confounder

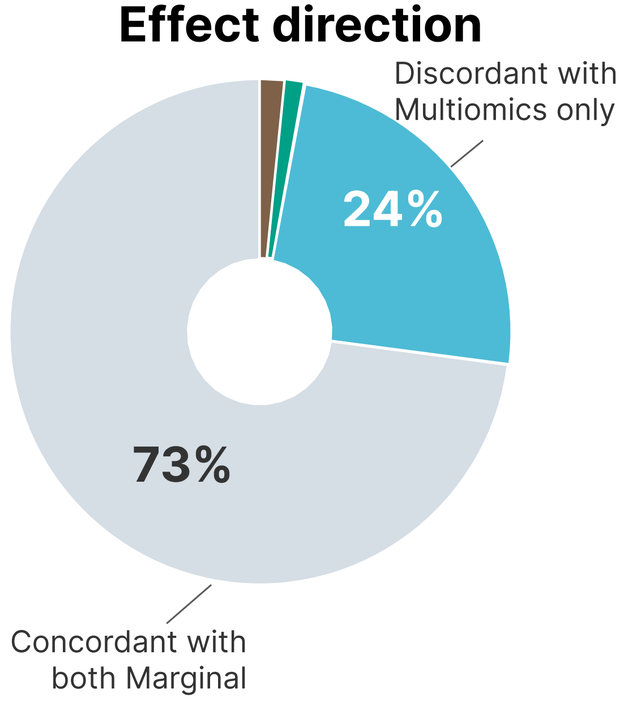

Correlation across individuals
Donors
D1 D2 D3 D4 D5
Gene g expression
D1 D2 D3 D4 D5
Peak p accessibility
NK
Gene g expression
Peak p accessibility


Correlation across cell types
Cell types
T B CD8 CD 4
Gene g expression
T B NK CD8 CD 4
Peak p accessibility
CD8 cells
CD4 cells
B cells
NK cells
T cells
Gene g expression
Peak p accessibility

B


CD4_Tn_CCR7
24632
6692
5693
1259
870
818
802
794
709
680
646
620
493
487
438
Intersection size
20000
10000
0
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
8627
9569
7265
10597
37181
17181
Significant pairs

C


Effect direction
Discordant with Multiomics only
24%
73%
Concordant with both Marginal